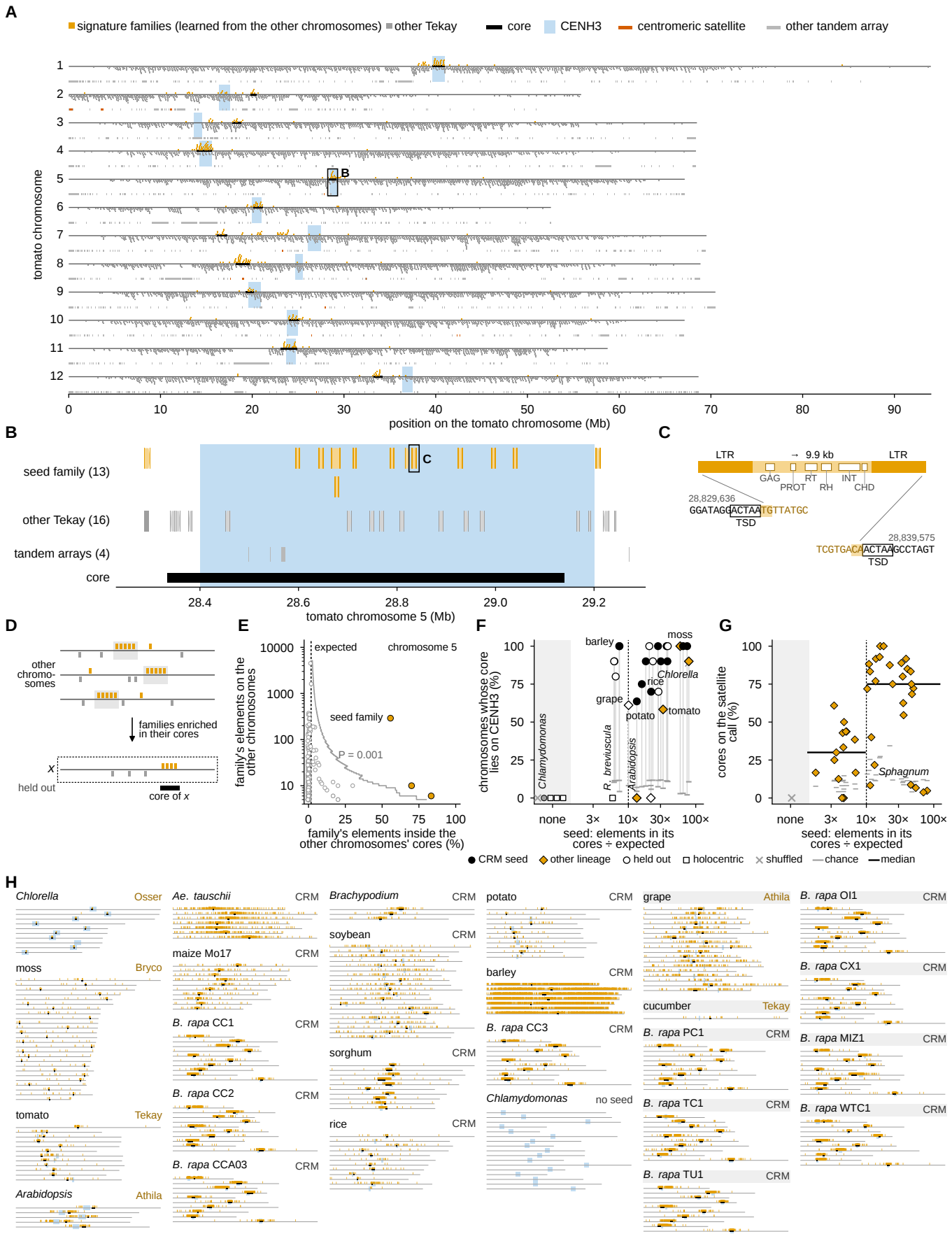

A
signature families (learned from the other chromosomes)
other Tekay
core
CENH3
centromeric satellite
other tandem array
tomato chromosome
position on the tomato chromosome (Mb)
B
seed family (13)
other Tekay (16)
tandem arrays (4)
core
tomato chromosome 5 (Mb)
C
LTR
9.9 kb
GAG
PROT
RT
RH
INT
CHD
28,829,636
GGATAGGACTAATGTTATGC
TSD
28,839,575
TCGTGACACACTAAAGCCTAGT
D
other chromo-somes
families enriched in their cores
held out
core of x
E
expected
chromosome 5
seed family
P = 0.001
family's elements on the other chromosomes
family's elements inside the other chromosomes' cores (%)
F
chromosomes whose core lies on CENH3 (%)
Chlamydomonas
R. brevuscula
Arabidopsis
barley
moss
Chlorella
rice
grape
potato
tomato
seed: elements in its cores ÷ expected
G
cores on the satellite call (%)
Sphagnum
seed: elements in its cores ÷ expected
CRM seed
other lineage
held out
holocentric
shuffled
chance
median
H
Chlorella
Osser
moss
Bryco
tomato
Tekay
Arabidopsis
Athila
Ae. tauschii
CRM
maize Mo17
B. rapa CC1
B. rapa CC2
B. rapa CCA03
Brachypodium
soybean
sorghum
rice
potato
barley
B. rapa CC3
Chlamydomonas
no seed
grape
Athila
cucumber
Tekay
B. rapa PC1
B. rapa TC1
B. rapa TU1
B. rapa OI1
B. rapa CX1
B. rapa MIZ1
B. rapa WTC1

Which LTR retrotransposons mark centromeres can be read from a genome alone: families that pile up at one locus on the other chromosomes mark each chromosome's centromere, in any lineage. The family, not the lineage, is the unit of centromere targeting.

A In tomato the Tekay lineage fills the pericentromeres, yet one family, 1 in 38 of its elements, piles up at one site per chromosome, on CENH3 in 7 of 12.

B Around one centromere, copies of that family crowd onto CENH3 among other Tekay copies.

C A typical copy is an intact insertion, with its target-site duplication and the chromodomain of its integrase.

D Each chromosome is left out in turn: the families concentrated in the cores of the other chromosomes define its signature, so no chromosome shapes its own.

E On a real chromosome, 3 families stand far above chance; the other 221 do not.

F Across plants and algae, founding families of CRM and of other lineages place the cores on CENH3 in most genomes; *Arabidopsis*' Athila cores flank it instead. Shuffled annotations give no seed; of 4 holocentric genomes, 3 give none and 1 a weak one whose cores sit mostly at chromosome ends.

G In polyploids without a CRM marker, families at least 10-fold specific mostly agree with independent satellite calls; less specific ones mostly do not.

H Every genome with truth, held-out genomes included: in most, the cores sit on CENH3.

Terms. Signature: the LTR retrotransposon families that mark one chromosome's centromere, learned from the other chromosomes. Core: the densest locus of the signature's elements on a chromosome (binomial test of the locus against its length, $P \le 0.001$). Seed: the lineage or family (CRM first) whose elements concentrate at one core on most chromosomes; its fold enrichment is the share of its elements inside its cores divided by the share of those chromosomes the cores cover. On CENH3: within the benchmark's tolerance of the CENH3 (CENP-A) ChIP interval (overlap in Arabidopsis thaliana, 0.1 Mb in moss, 1 Mb in tomato, 8 Mb in barley). Family labels come from one LTRquest run per genome and mean nothing across genomes.

A All 12 tomato chromosomes at one scale; every Tekay element (11,657) is a glyph at its midpoint, stacked where glyphs crowd (0.3 mm slots). Above each chromosome, elements of the families recruited for it (3 families over the genome; the seed family has 313 elements, 308 of them classed Tekay: 1 in 38 Tekay elements); below, the other Tekay elements. Black bar: the chromosome's core; blue: CENH3; beneath, every tandem array (6,917; vermillion where the satellite line classed it centromeric). The lineage fills tens of megabases; the family piles at one site. Cores lie on CENH3 on 7 of 12 chromosomes (seed 32.7-fold); they miss on chromosomes 2, 3, 7, 8, 12, all drawn. With these LTRs alone cenfuse 0.12 places 6 of 12 centromeres right (0.11: 1); on 2 of the misses (8, 12) its call stays medium; with Hi-C, 10 of 12.

B Chromosome 5, 28.23-29.30 Mb (box in A): every Tekay element in the window drawn to size, as an arrow in its orientation with its two LTRs dark; seed family above, other Tekay below (2 other recruited families have no copy here), then the tandem arrays. The core (black), the densest locus of the recruited families, overlaps CENH3. Example rule: among chromosomes whose core lies on CENH3, the one whose signature-element count is nearest the genome median (22.5); 4 tied (5, 6, 10, 11), resolved by the count of all Tekay elements nearest its median (1047; chromosome 5: 1051).

C The boxed element (LTRRT_08819, 9,940 bp; family median 9,709 bp): its LTRs, protein domains in Tekay order (GAG-PROT-RT-RH-INT-CHD; CHD, the chromodomain that steers chromovirus integration) and both ends read from the genome: the 5-bp target-site duplication (ACTAA) on each side and the TG...CA ends of the LTRs. Chosen as the element nearest the family's median length among the 5 of 11 seed-family elements in this core that pass these checks (high-confidence call, exact TSD, TG...CA, both LTRs, domains in order).

D Recruitment, leave one chromosome out. For chromosome *x*, every family with at least 5 elements on the other chromosomes is tested: are more of its elements inside those chromosomes' cores than the share of their length the cores cover (binomial, Bonferroni over the families tested, $P \le 0.001$)? The families that pass make *x*'s signature, and their densest locus on *x* is its core. Rounds repeat from the new cores until they stop changing. A family recruited elsewhere that also concentrates on *x* is independent evidence there.

E The test of D for chromosome 5, last round: each circle is one of the 224 families tested on the other 11 chromosomes. Dashed: the share expected by chance (1.7%, the share of those chromosomes the cores cover); gray step: the share needed for $P = 0.001$. The 3 recruited families (filled; lineage Tekay) include the seed family, 162 of whose 291 elements on the other chromosomes lie inside their cores ($P = 5e\text{-}199$).

F One point per genome with LTRs and truth: the seed's fold enrichment against the share of chromosomes whose core lies on CENH3; the gray dash below each is the share expected if each core were placed at random on its chromosome. Development set: 16 genomes with a seed (circles CRM; diamonds other lineages: moss Bryco 58.8-fold, *Chlorella* Osser 78.9, tomato Tekay 32.7, *Arabidopsis* Athila 13.3). Open, held out from the 0.12 rules (cores on CENH3): *B. rapa* PC1 9/10, *B. rapa* TC1 9/10, *B. rapa* TU1 10/10, *B. rapa* OI1 8/10, *B. rapa* CX1 10/10, *B. rapa* MIZ1 7/10, *B. rapa* WTC1 9/10, cucumber 0/1, grape 11/18. Gray column: genomes and controls without a seed, hence without cores: *Chlamydomonas*; the 51 annotations of the 17 genomes with element positions shuffled within chromosomes (3 per genome; 0 of all 157 shuffled annotations, polyploids included, give a seed); and, as open squares, the holocentric genomes held out with LTRs: *C. elegans*, *B. mori*, *C. nigoni* give no seed (0 confident calls). *R. breviuscula*, also holocentric and without CENH3 truth, is drawn at its fold: an Athila seed at 5.8-fold with significant cores on 5 of 5 chromosomes, 4 of them within 1 Mb of a chromosome end; under 10-fold it only flags (LTRs alone: 0 confident calls). Under the frozen rules, Hi-C anchored on those cores passed through the clustering of chromosome ends (3 medium calls with Hi-C); the rule was changed after this result, so that an anchored peak counts only where it is clean (not at a chromosome end, hub or NOR). Dashed: 10-fold, the level from which a lineage other than CRM may decide. *Arabidopsis*' Athila cores lie 0.46-1.38 Mb beside the CENH3 interval on all 5 chromosomes, scored by overlap; Athila only flags there, because the CEN180 satellite passes the one-per-chromosome test (as in held-out grape). With these LTRs alone, 0.12 is right on moss 26/26 (0.11: 16), barley 7/7 (4), potato 10/11 (6), *Chlorella* 9/10 (8), rice 12/12, *Arabidopsis* 5/5; over all 324 chromosomes with LTRs and truth, 248 to 292 right and 29 to 12 confidently wrong.

G The origin of the 10-fold level (development measurement, no truth): 42 polyploids whose seed is not CRM; y, the share of chromosomes whose core lies within 1 Mb of the 0.11 FASTA-only call, counting only calls that engine did not place from retrotransposon clusters (91 of 934 excluded); gray dashes, chance; gray column, 106 shuffled polyploid annotations, no seed. Median 0.30 at or below 10-fold ($n = 15$) against 0.75 above ($n = 27$; one-sided Mann-Whitney $P = 0.0004$); with all calls, 0.30 and 0.75. Exceptions above 10-fold: three *Sphagnum* genomes (Bryco, 54-80-fold) agree on only 0.04-0.07 of such calls (0.32-0.63 counting those placed from retrotransposon clusters), and so do *Lotus subbiflorus*, *Malus toringoides*, *Solanum stoloniferum*. Satellite calls are not truth: tomato's satellites are wrong where its Tekay cores are right.

H Every genome with LTRs and truth, one row per chromosome with truth, each genome at its own scale: signature elements as ticks, the significant core black, CENH3 blue; top right, the seed's lineage. Gray title band: the genomes planned as held out from the 0.12 rules (grape, cucumber, PC1, TC1, TU1, OI1, CX1, MIZ1, WTC1), laid out in full whether scored or not; all were scored when this figure was drawn. The 17 other genomes are the development set that shaped the rules.

Exceptions. Tomato's cores miss CENH3 on 5 of 12 chromosomes (A); *Arabidopsis*' Athila cores flank it (F); in the holocentric *R. breviuscula* an Athila seed under 10-fold puts its cores at chromosome ends (F). A core of a lineage other than CRM that holds under half of the signature's elements on its chromosome only flags. The 10-fold level and the rules were set on the development set; held-out genomes were scored once with the frozen code.

Analysis: asc-oulab:/data2/chris/centromere_benchmark/v2/v3/v4/v5/v6/v7/v8/v9/v10/v10.1/v10.2/v10.3/v10.4/v11 · Script: /data2/chris/centromere_benchmark/v2/v3/v4/v5/v6/v7/v8/v9/v10/v10.1/v10.2/v10.3/v10.4/v11/figures/fig_ltr_signature/fig_ltr_signature.py · 2026-11-07